Trapezoidal Method Part 1

Monday, 23 March 2026 9:13 am

Calculation of Bulk Volume (BV) Using Isopach Map – Trapezoidal Method

The isopach map shows lines of equal reservoir thickness. It is used to determine the bulk rock volume (BV) of the reservoir. Bulk volume is calculated by integrating the areas between contour lines multiplied by the thickness interval.

$$BV = \left(\frac{h}{2}\right) x(A_0 + 2A_1 + 2A_2 + \cdots + 2A_{n-1} + A_n)$$

C.L	Area *inch²*
0	A_0 WOC
10	A_1
20	A_2
30	A_3
40	A_4
	GOC A'
50	A_5
60	A_6
70	A_7
76	O

Note:
C.L refers to contour level (thickness in ft).
Area represents the area enclosed by each contour.
Areas are measured using planimeters, grid method, or GIS tools.
The contour interval, h, is typically constant (example: 10 ft).
In some cases, a partial thickness (h') occurs when the fluid contact (e.g., GOC) lies between two contours.

BV = Area x Thickness
(Net pay)

(net reservoir volume)

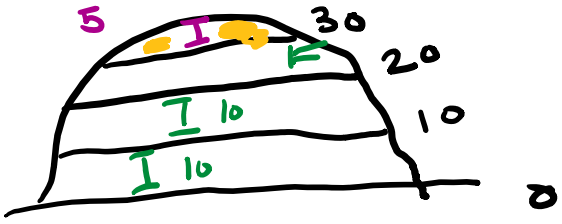
Contour maps
- Structural maps → contour Levels
- Isopach maps → areas

Trapezoidal Method Concept

The trapezoidal method approximates the volume between two contours as a trapezoid. The volume between contours was as follows:

$$V = \left(\frac{h}{2}\right) x(A_i + A_{i+1})$$

Where:
h = contour interval
A_i = lower contour area
A_{i+1} = upper contour area



The Bulk Volume Formula
The total bulk volume is obtained by summing all trapezoidal segments.

$$BV = \left(\frac{h}{2}\right) x(A_0 + 2A_1 + 2A_2 + \cdots + 2A_{n-1} + A_n)$$

Where:

h = contour interval
A_n = nth contour area

Adjustment for Gas-Oil Contact
When the GOC occurs between contour levels, interpolation is required. The additional volume is calculated as follows:

$$\left(\frac{h'}{2}\right)(A_n + A')$$

Where:

h' = partial thickness

A' = interpolated area at GOC

Hence, the Final Bulk Volume Equation:

$$BV = \left(\frac{h}{2}\right) x(A_0 + 2A_1 + 2A_2 + \cdots + 2A_{n-1} + A_n) + \left(\frac{h'}{2}\right)(A_n + A')$$

Condition:

$$\frac{A_n}{A_{n-1}} > 0.5$$

Symbol	Meaning
(A_0)	Area at the lowest contour (usually WOC)
(A_1, A_2, ...)	Areas at intermediate contour levels
(A_n)	Area at the last full contour before the contact
(A')	Area at the interpolated contact level (GOC or other)
(h)	Regular contour interval
(h')	Partial interval

Trapezoidal Method Part 2

Monday, 23 March 2026 9:15 am

Sample Problem

Given the following planimetred areas of an oit of reservoir. Calculate the original oil place (N) if φ =25%, S_{wi}=30%, β_{oi}=1.4 bbl/STB and map scale=1:15000

C.L :	0	10	20	30	40	50	60	70	80	86
Area inch² :	250	200	140	98	76	40	26	12	5	0

Given:

Porosity = 25% = 0.25

Initial water saturation = 30% = 0.3

Initial Oil formation volume factor = 1.4 bbl / STB

Map scale: 1:15000

Conversion of Map Area to Acres

Scale	1 inch = 15000 inches
Area Scale	$1\ in^2 = (15000)^2\ in^2$
Conversion of ft² to in²	$1ft^2 = 144\ \text{in}^2$
Conversion of acre to ft²	$1acre = 43560ft^2$
Conversion of in² to acres	$1in^2 = 35.85\ acres$

Contour Level	**Area (in²)**	**Area (acres)**
0	250	8962.5
10	200	7170
20	140	5019
30	98	3513
40	76	2725
50	40	1434
60	26	932
70	12	430
80	5	179
86	0	0

Use this formula

$$BV = \left(\frac{h}{2}\right) x(A_0 + 2A_1 + 2A_2 + \cdots + 2A_{n-1} + A_n) + \left(\frac{h'}{2}\right)(A_n + A')$$

Ao = 8962.5 acres
h = 10 ft
h' = 6 ft
An = 179 acres
A' = 0 acres

Solution:

$$BV_{Total} = BV_1 + BV_2$$

$$BV = \left(\frac{h}{2}\right) x(A_0 + 2A_1 + 2A_2 + \cdots + 2A_{n-1} + A_n) + \left(\frac{h'}{2}\right)(A_n + A')$$

$$BV_1 = \left(\frac{h}{2}\right)[A_o + 2(A_1 + A_2 + A_3 + A_4 + A_5 + A_6 + A_7) + A_8]$$

$$= \left(\frac{10}{2}\right)[8962.5 + 2(7170 + 5019 + 3513 + 2725 + 1434 + 932 + 430) + 179]$$

$$BV_1 = 257937.5\ acre - ft$$

$$BV_2 = \left(\frac{h'}{2}\right)(A_n + A') = \left(\frac{6}{2}\right)(179 + 0)$$

$$BV_2 = 537\ acre - ft$$

$$BV_{Total} = BV_1 + BV_2 = 257937.5 + 573$$

$$BV = 258937.5\ acre - ft$$

Original Oil in Place

$$N = \frac{7758\ x\ A\ x\ h\ x\ \Phi\ x\ (1 - S_{wi})}{B_{oi}}$$

$$N = \frac{7758\ x\ BV\ x\ \Phi\ x\ (1 - S_{wi})}{B_{oi}}$$

$$N = \frac{7758 \times 258937.5 \times 0.25 \times 0.70}{1.4} \approx 2.51 \times 10^8$$

Note:

$$N = \frac{}{1.4} \approx 2.51 \times 10$$

Note:

Bulk Volume = 259369.5 $acres - ft$

Porosity = 25% = 0.25

Initial water saturation = 30% = 0.3

Initial Oil formation volume factor = 1.4 bbl / STB

Final Answer:

$$\boxed{OOIP \approx 2.51 \times 10^8\ STB \approx 251\ MMSTB}$$

Pyramidal Method

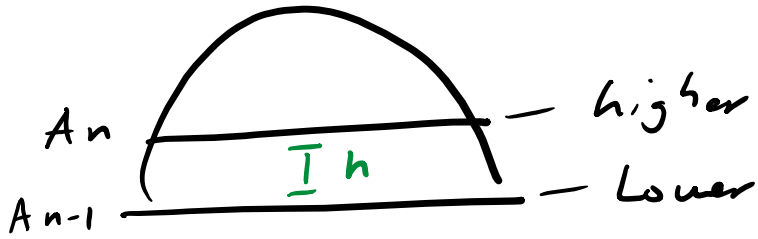
Monday, 23 March 2026 9:17 am

Calculation of Bulk Volume (BV) Using Isopach Map – Pyramid Method

$$BV = \frac{h}{3}\left(A_{n-1} + A_n + \sqrt{A_{n-1}A_n}\right)$$

Condition:

$$\frac{A_n}{A_{n-1}} \leq 0.5$$



Sample Problem (Modified)

Given the following planimetred areas of an oit of reservoir. Calculate the original oil place (N) if φ =25%, S_{wi}=30%, β_{oi}=1.4 bbl/STB and map scale=1:15000

C.L :	0	10	20	30	40	50	60	70	80	86
Area inch² :	250	200	140	98	76	40	26	12	5	0

Given:
Porosity = 25% = 0.25
Initial water saturation = 30% = 0.3
Initial Oil formation volume factor = 1.4 bbl / STB
Map scale: 1:15000

Conversion of Map Area to Acres

Scale	1 inch = 15000 inches
Area Scale	$1\ in^2 = (15000)^2\ in^2$
Conversion of ft² to in²	$1ft^2 = 144\ \text{in}^2$
Conversion of acre to ft²	$1acre = 43560ft^2$
Conversion of in² to acres	$1in^2 = 35.85\ acres$

$$\frac{A_n}{A_{n-1}} \leq 0.5$$

Contour Level	Area (in²)	Area (acres)
0	250	8962.5
10	200	7170
20	140	5019
30	98	3513
40	76	2725
50	40	1434
60	26	932
70	12	430
80	5	179
86	0	0

Contour Level Interval	A_n/A_{n-1}	$\frac{A_n}{A_{n-1}} \leq 0.5$	Interpretation
0–10	0.80	No	Use trapezoidal
10–20	0.70	No	Use trapezoidal
20–30	0.70	No	Use trapezoidal
30–40	0.78	No	Use trapezoidal
40–50	0.53	No	Use trapezoidal
50–60	0.65	No	Use trapezoidal
60–70	0.46	Yes	Use Pyramidal
70–80	0.42	Yes	Use Pyramidal

50–60	0.65	No	Use Trapezoidal
60–70	0.46	Yes	Use Pyramidal
70–80	0.42	Yes	Use Pyramidal
80–86	0.00	Yes	Use Pyramidal

Solution:

Using Trapezoidal Method (A_0-A_6):

$$BV_1 = \left(\frac{h}{2}\right)[A_o + 2(A_1 + A_2 + A_3 + A_4 + A_5) + A_6]$$

$$= \left(\frac{10}{2}\right)[8962.5 + 2(7170 + 5019 + 3513 + 2725 + 1434) + 932]$$

$$BV_1 = 248082.5\ acre - ft$$

Using Pyramidal Method (A_6-A_9)

$$BV_2 = \frac{h}{3}[A_6 + A_7 + \sqrt{A_6A_7}] + \frac{h}{3}[A_7 + A_8 + \sqrt{A_7A_8}] + \frac{h'}{3}[A_8 + A_9 + \sqrt{A_8A_9}]$$

$$BV_2 = \left(\frac{10}{3}\right)[932 + 430 + \sqrt{(932)(430)}] + \left(\frac{10}{3}\right)[430 + 179 + \sqrt{(430)(179)}] + \left(\frac{6}{3}\right)[179 + 0 + \sqrt{(179)(0)}]$$

$$BV_2 \approx 9962.97\ acre - ft$$

Bulk Volume:

$$BV_{Total} = BV_1 + BV_2 = 248082.5 + 9962.97$$

$$BV_{Total} \approx 258044\ acre - ft$$

Original Oil in Place

$$N = \frac{7758\ x\ A\ x\ h\ x\ \Phi\ x\ (1 - S_{wi})}{B_{oi}}$$

$$N = \frac{7758\ x\ BV\ x\ \Phi\ x\ (1 - S_{wi})}{B_{oi}}$$

$$N = \frac{7758 \times 258044 \times 0.25 \times 0.70}{1.4} \approx 2.50 \times 10^8$$

$$OOIP \approx 2.50 \times 10^8\ STB \approx 250\ MMSTB$$

Note:
Bulk Volume = 258044 $acres - ft$
Porosity = 25% = 0.25
Initial water saturation = 30% = 0.3
Initial Oil formation volume factor = 1.4 bbl / STB

Calculation of Bulk Volume (BV) Using Isopach Map – Simpson Method

$$BV = \frac{h}{3}(A_0 + 4A_1 + 2A_2 + 4A_3 + \cdots + 2A_{n-2} + 4A_{n-1} + A_n)$$

Condition: Odd number of Contour lines

Odd-numbered Area multiplied by 4

Even-numbered Area multiplied by 2

Condition
1. number of contour line = odd
2. thickness of contour interval => uniform

Sample Problem (Modified)

Given the following planimetred areas of an oit of reservoir. Calculate the original oil place (N) if φ =25%, S_{wi}=30%, β_{oi}=1.4 bbl/STB and map scale=1:15000

C.L :	0	10	20	30	40	50	60	70	80	86
Area inch² :	250	200	140	98	76	40	26	12	5	0

10 10 10 10 10 10 10 10

T, P

Given:
Porosity = 25% = 0.25
Initial water saturation = 30% = 0.3
Initial Oil formation volume factor = 1.4 bbl / STB
Map scale: 1:15000

Conversion of Map Area to Acres

Scale	1 inch = 15000 inches
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Conversion of ft² to in²	$1ft^2 = 144\ \text{in}^2$
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80	5	179
86	0	0

Simpson

3 Pyramid

$$\frac{A_n}{A_{n-1}} \leq 0.5$$

$$\frac{0}{5} \leq 0.5$$

Analysis:

Based on the available data, the total number of contour lines is ten (10). However, Simpson's Method requires an odd number of contour lines (or an even number of intervals) for proper application. As such, the method is not directly applicable in its current form.

To address this limitation, the last contour line (C.L. 86) is excluded from the analysis. Simpson's Method is then applied to the remaining contour levels, ranging from 0 to 80.

With this adjustment, the number of contour lines is reduced to nine (9), which satisfies the requirement of an odd number of contour lines. Therefore, Simpson's Method can be validly applied to the modified dataset.

Solution:

Using Simpson Method:

$$BV_1 = \frac{10}{3}[250 + 4(200) + 2(140) + 4(98) + 2(76) + 4(40) + 2(26) + 4(12) + 5]$$

h, A_0, A_1, A_2, A_3, A_4, A_5, A_6, A_7, $A_n = A_8$

$$BV_1 = 7130\ in^2 - ft$$

Using Triangular Method:

Using Triangular Method:

$$BV_2 = \frac{6}{3}\left[5 + 0 + \sqrt{(5)(0)}\right]$$

$$BV_2 = 10\ in^2 - ft$$

Bulk Volume:

$$BV_{total} = BV_1 + BV_2 = 7130 + 10 = 7140\ in - ft$$

$$BV_{total} = 7140\ in^2 - ft \approx 255969\ acre - ft$$

Original Oil in Place

$$N = \frac{7758\ x\ A\ x\ h\ x\ \Phi\ x\ (1 - S_{wi})}{B_{oi}}$$

$$N = \frac{7758\ x\ BV\ x\ \Phi\ x\ (1 - S_{wi})}{B_{oi}}$$

$$N = \frac{7758 \times 255969 \times 0.25 \times 0.70}{1.4} \approx 2.48 \times 10^8$$

$$OOIP \approx 2.48 \times 10^8\ STB \approx 248\ MMSTB$$

Note:
Bulk Volume = 259369.5 $acres - ft$
Porosity = 25% = 0.25
Initial water saturation = 30% = 0.3
Initial Oil formation volume factor = 1.4 bbl / STB

A gas cap has the following data : φ =25%, S_{wi}=30%, β_{oi}=1.3 bbl/STB, β_{gi}=0.001 bbl/SCF and map scale=1:20000

C.L :	0(WOC)	10	20	30	33(GOC)	40	50	60	70	76
Area inch^2 :	350	310	270	220	200	190	130	55	25	0

Calculate the original oil in place (N) and the original gas in place (G)